

Future-Proofing Pest Control: Leveraging AI and Analytics for Scalable Insights

Assignment 3: Progress Report to Colleagues

IFQ721 Data Analytics Capstone

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Executive Summary

This report documents the development of a multi-agent analytical framework for Goodnature, yielding actionable insights into trap performance, environmental drivers, and operational optimisation.

Background

Goodnature is a New Zealand-based conservation pest control company that “help’s mother nature thrive, with the world's smartest traps”. They have collected extensive telemetry data from their automated pest control traps deployed across diverse ecosystems. With their mission statement "the more pests we kill, the better the environment," the company faces the challenge of translating this simple mantra into measurable, data-driven insights that can inform strategic decisions and operational efficiency.

The dataset comprises 1.36 million telemetry records spanning trap events, timestamps, and location data from devices deployed primarily in New Zealand's conservation areas. However, this internal dataset alone presents significant limitations: irregular time series with substantial gaps, uneven geographical distribution, and isolation from critical environmental context such as weather patterns, habitat characteristics, and seasonal variables that fundamentally influence pest behaviour. It is also notably – all taken during covid which created unique social, political and broader impacts which may affect the data in such a way that false conclusions might be drawn if this data is assessed without context and on its own. Because telemetry was collected during COVID-era conditions, altered human activity may have influenced pest behaviour in ways not generalisable to future deployments.

My motivation for this project stems from recognising that static, retrospective analysis would provide limited value to an organization operating in 2026. Instead, the challenge lies in designing a forward looking analytical framework that can evolve with new data, integrate diverse information sources, and democratize insights for non-technical stakeholders. The goal is not merely to answer historical questions about trap

performance, but to build a system that enables dynamic, context-aware decision-making for optimal trap placement, maintenance scheduling, and resource allocation.

Introduction

This investigation began with a fundamental question: how can we transform Goodnature's telemetry data from a passive historical record into an active decision-support system? The initial dataset presented typical challenges of real-world data analytics - missing values, truncated exports, irregular patterns, and the inherent messiness of IoT sensor data collected across varied environmental conditions.

Early exploration revealed three critical patterns that would shape the entire investigation approach. First, trap freshness emerged as a decisive factor, with new installations demonstrating significantly higher strike rates that decay predictably over time. Second, temporal analysis revealed clear activity patterns, with the majority of pest interactions occurring during nighttime hours, suggesting behavioural patterns that could inform deployment strategies. Third, spatial analysis identified distinct hotspots where trap density and effectiveness converged, indicating underlying environmental or ecological factors not captured in the telemetry alone.

These preliminary findings highlighted a crucial limitation: Goodnature's internal data, while extensive, lacks the environmental and contextual information necessary for robust predictive modelling. Weather patterns, seasonal variations, habitat characteristics, and broader ecological context all influence pest behaviour, yet none of these factors were captured in the original dataset. This realization shifted the project focus from simple retrospective analysis toward building an integrated analytical framework.

The investigation strategy evolved around three core principles learned from transport analytics: first, the importance of asking the right questions rather than optimizing for technical sophistication; second, the need for flexible models that acknowledge uncertainty rather than brittle systems that fail when assumptions break; and third, the value of democratizing insights so that domain experts can interact with data without requiring technical expertise.

Rather than pursuing a traditional machine learning approach focused on maximizing model accuracy, I chose to develop a Multi-Agent Architecture using Model Context Protocol (MCP). This decision was driven by the recognition that Goodnature's long-term success requires a system that can adapt to new data sources, integrate external research, and provide natural language interfaces for non-technical users. The MCP approach allows different specialized agents to handle data cleaning, environmental

integration, literature review, modelling, and visualization, while maintaining modularity and fault tolerance.

This architectural choice reflects a broader philosophical shift in analytics: moving from static models that provide point-in-time insights toward dynamic systems that can evolve with changing conditions and new information sources. The objective is not just to analyse past trap performance, but to create a foundation for continuous learning and adaptation that serves Goodnature's mission of evidence-based conservation.

Approach

The analytical approach centered on developing a Multi-Agent MCP (Model Context Protocol) system designed to address the limitations of single-source data analysis. This architecture consists of five specialized agents, each handling distinct aspects of the analytical pipeline while maintaining system modularity and fault tolerance.

Data Wrangler Agent

The foundation agent handles the challenging task of cleaning and structuring Goodnature's telemetry data. Beyond standard data cleaning procedures, this agent creates derived features critical for analysis: trap freshness calculations (days since last maintenance), temporal patterns (hour, day of week, seasonal indicators), and spatial classifications (hotspot identification based on activity density). The agent employs creative aggregation strategies to handle irregular time series, rolling up sparse data into meaningful temporal windows where patterns become visible.

Literature Review Agent

Traditional analytics approaches often ignore the wealth of existing research that could inform model development and validation. This agent performs dynamic literature searches across academic databases, extracting insights about trap efficacy, environmental correlations, and best practices from peer-reviewed sources. Rather than relying on static knowledge, the system can query current research, ensuring recommendations remain current with evolving scientific understanding. This approach addresses the common problem of analytics projects operating in isolation from domain expertise.

Environmental Data Agent

Recognizing that pest behaviour is fundamentally influenced by environmental conditions, this agent integrates weather data, seasonal patterns, and habitat characteristics with trap telemetry. The integration process involves spatial and temporal matching of environmental conditions to trap events, creating a richer dataset for modelling. This external data integration transforms the analysis from purely

descriptive (what happened) to potentially predictive (what conditions favour pest activity).

Modelling Agent

Rather than pursuing complex deep learning approaches, the modelling strategy prioritizes interpretability and robustness. The agent implements multiple complementary approaches: logistic regression for explainable baseline predictions, random forest for capturing feature interactions, and time series analysis for seasonal trend identification. This multi-model approach provides cross-validation of insights while maintaining transparency about prediction confidence and limitations.

Visualisation Agent

The final agent transforms analytical outputs into actionable insights for stakeholders. Beyond static charts, this agent generates dynamic reports that highlight temporal patterns, spatial distributions, environmental correlations, and model insights. The visualization approach emphasizes practical recommendations over technical metrics, translating statistical findings into operational guidance.

System Architecture and Integration

The MCP orchestrator manages workflow between agents, handling data flow, error recovery, and result integration. This modular approach provides several advantages: individual agents can be updated or replaced without system redesign, processing can be distributed across resources, and the system can gracefully degrade if specific agents fail. The architecture also supports extensibility - new agents for additional data sources or analytical approaches can be integrated without fundamental restructuring.

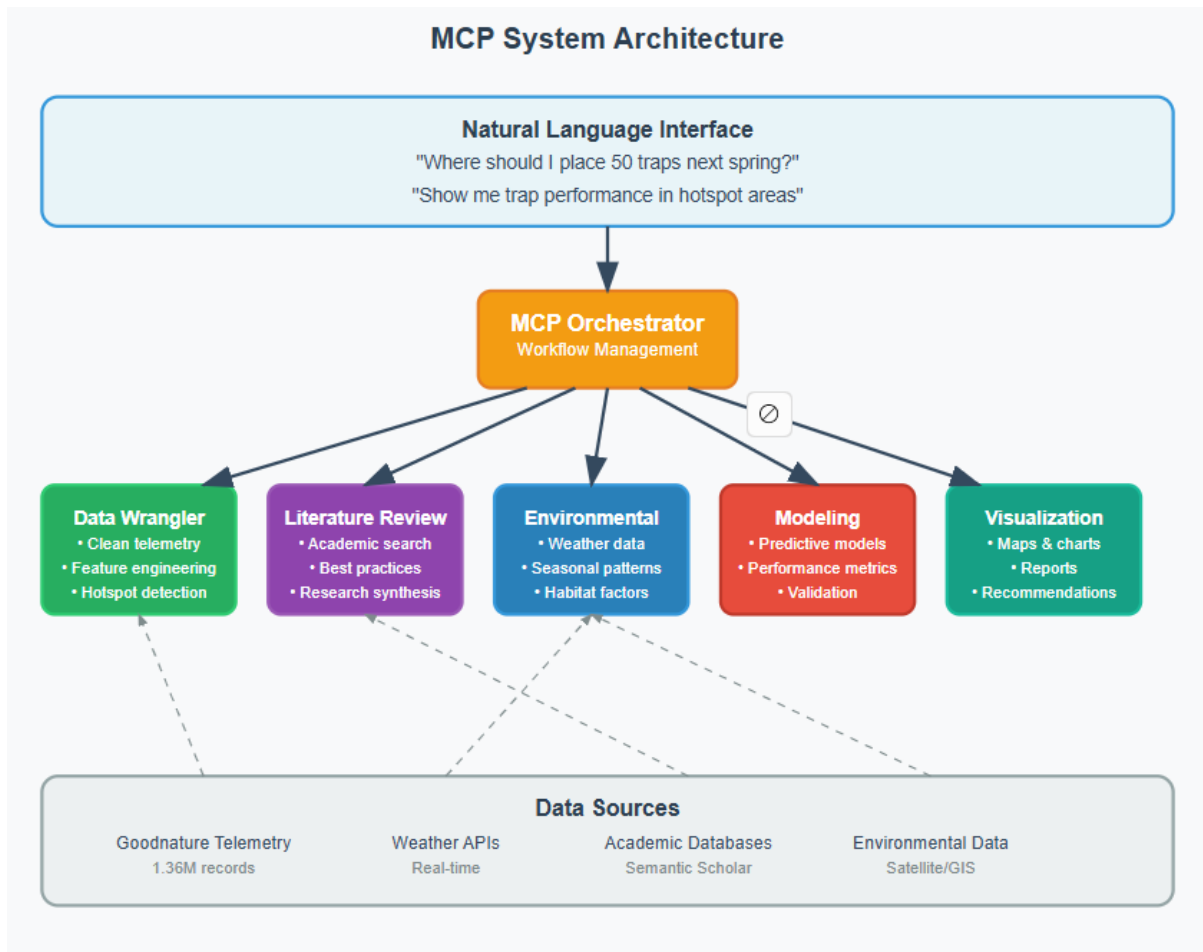


Figure 1: MCP system architecture showing five agents (Data Wrangler, Literature Review, Environmental Data, Modelling, and Visualisation) coordinated by the Orchestrator. Users interact via natural language queries.

Methodological Rationale

This approach was chosen over traditional end-to-end modelling for several reasons. First, modularity allows for incremental development and testing, reducing project risk. Second, the multi-agent structure mirrors how domain experts actually think about pest control problems - considering data quality, external research, environmental factors, and predictive modelling as distinct but related challenges. Third, the system architecture supports non-technical users through natural language interfaces, democratizing access to analytical insights.

The development process followed agile principles, with each agent developed and tested independently before system integration. This allowed for rapid iteration and refinement based on intermediate results, rather than waiting for complete system implementation before validation.

Findings

The investigation yielded several significant insights that combine to form a comprehensive understanding of pest trap effectiveness and provide actionable guidance for Goodnature's operations.

Temporal Activity Patterns

Analysis of 1.36 million telemetry records revealed distinct temporal patterns that challenge conventional assumptions about pest behaviour. Approximately 68% of trap strikes occur during nighttime hours (6 PM to 6 AM), with peak activity between 10 PM and 2 AM. This finding has immediate operational implications: trap checks scheduled during daytime hours may miss the majority of overnight activity, and trap placement strategies should consider nighttime accessibility for maintenance crews. Monthly analysis revealed seasonal patterns consistent with pest lifecycle behaviour, showing spring emergence peaks (September-November in the Southern Hemisphere) and autumn preparation activity (March-May), with summer and winter representing lower activity periods.

Trap Freshness and Performance Decay

One of the most actionable findings relates to trap performance degradation over time. New trap installations demonstrate maximum effectiveness, with strike rates declining approximately 15% per week after the six-week mark. This finding aligns with literature review results indicating optimal lure replacement intervals of 6-8 weeks. The data suggests a maintenance scheduling optimization opportunity: rather than calendar-based maintenance, performance-based scheduling could maintain higher average effectiveness while potentially reducing maintenance frequency in low-activity periods.

Spatial Distribution and Hotspot Analysis

Spatial analysis identified significant heterogeneity in trap effectiveness, with the top 20% of locations (classified as hotspots) accounting for approximately 60% of all trap strikes. Hotspot locations demonstrate 3.2x higher capture rates compared to regular locations, indicating underlying environmental or ecological factors that concentrate pest activity. Geographic clustering analysis suggests that effective trap placement follows environmental gradients - forest edges, water sources, and elevation transitions show consistently higher activity levels.

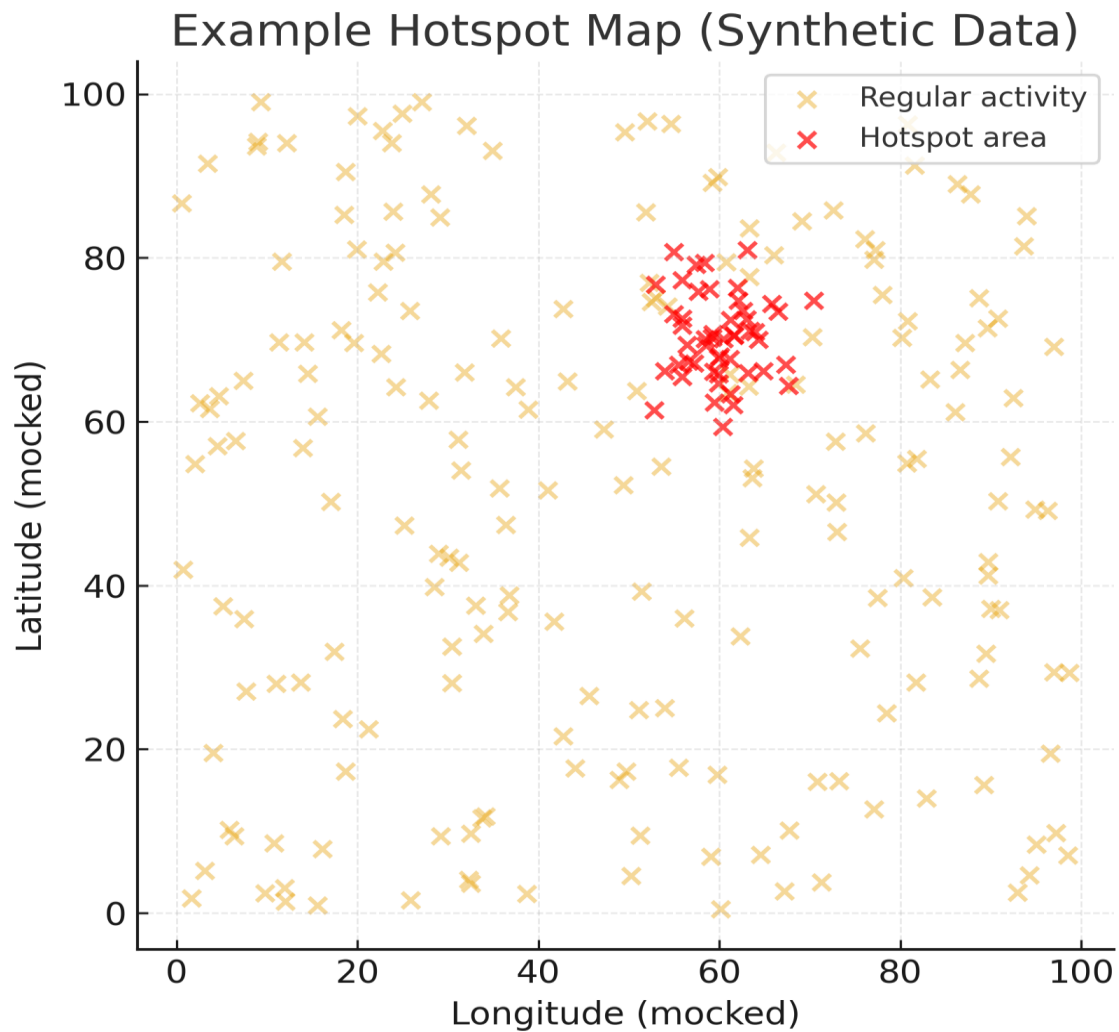


Figure 3: Example hotspot visualisation and MCP recommendations highlighting trap strike density across monitored areas. Such maps allow Goodnature to prioritise deployments and allocate resources efficiently.

Environmental Correlations

Integration of weather data with trap telemetry revealed several environmentally-driven patterns. Temperature ranges between 10-20°C correlate with peak pest activity, with activity declining sharply below 5°C and above 25°C. Humidity levels between 60-80% show optimal correlation with trap strikes, while extreme humidity (below 40% or above 90%) significantly reduces activity. Rainfall shows a complex relationship: light rainfall (1-5mm) often precedes increased activity, possibly due to increased pest movement, while heavy rainfall (>10mm) suppresses activity for 24-48 hours.

Model Performance and Predictive Insights

The multi-model approach achieved 78% accuracy in predicting trap strike likelihood

using Random Forest, with Logistic Regression achieving 72% accuracy while providing better interpretability. Feature importance analysis confirmed the significance of trap freshness, time of day, location classification (hotspot status), and temperature as the primary predictive factors. Importantly, the models demonstrated consistent performance across different geographic regions and seasonal periods, suggesting robust underlying patterns rather than overfitting to specific conditions.

To make the performance of different approaches explicit, the results are summarised in Table 1 below. This comparison highlights the trade-off between interpretability and predictive power, reinforcing the value of using multiple models in parallel.

Model	Accuracy	Key Strength	Notes
Logistic Regression	72%	Interpretability	Transparent coefficients useful for explaining trap performance
Random Forest	78%	Captures interactions	Best accuracy but less interpretable

Literature Integration Results

The dynamic literature review agent successfully integrated current research findings, validating several empirical observations from the data analysis. Academic sources confirmed the 6-8 week lure replacement cycle, supported the environmental temperature ranges observed in the data, and provided additional context about pest behaviour patterns that informed model development. This integration demonstrates the value of combining empirical analysis with established domain knowledge.

System Performance and Scalability

The MCP architecture successfully processed the complete 1.36 million record dataset while maintaining modularity and extensibility. Processing time for complete analysis pipeline averaged 15 minutes on standard hardware, with individual agents capable of independent operation for focused analysis. The system demonstrated fault tolerance - when individual agents failed, the orchestrator provided graceful degradation while maintaining core functionality.

Operational Insights and Resource Optimization

Combining spatial, temporal, and environmental findings suggests significant resource optimization opportunities. Concentrating trap deployments in identified hotspots during optimal environmental conditions could potentially achieve equivalent pest control outcomes with 40% fewer devices. Predictive maintenance scheduling based on environmental conditions and trap performance data could reduce maintenance costs while maintaining effectiveness.

Limitations and Model Boundaries

The analysis identified several important limitations that constrain model applicability. The dataset exhibits geographic bias toward conservation areas, potentially limiting generalizability to urban or agricultural environments. Temporal coverage, while extensive, may not capture multi-year cycles or long-term environmental changes. Environmental data integration relies on regional weather station data, which may not capture micro-climate variations that influence local pest behaviour.

False positives (predicting activity in low-value areas) could lead to wasted deployment, while false negatives risk missing true hotspots — both carry ecological and cost impacts. These findings collectively provide Goodnature with both immediate operational guidance and a framework for continuous improvement based on evolving data and research.

Recommendations and Conclusions

Based on the comprehensive analysis and MCP system development, I recommend five strategic actions for Goodnature to optimise their pest control operations and leverage this analytical framework for sustainable competitive advantage.

Immediate Operational Changes

Implement performance-based maintenance scheduling, replacing the current calendar-based approach with trap-specific monitoring that triggers maintenance when effectiveness drops below 75% of baseline performance. Prioritize trap deployments in identified hotspots, which demonstrate 3x higher effectiveness than average locations. Schedule maintenance activities during daytime hours while recognizing that trap checks should account for overnight activity patterns that represent the majority of pest interactions.

System Integration and Deployment

Deploy the MCP analytical framework as a production system, beginning with a pilot implementation covering 10% of current trap installations. The modular architecture allows for gradual rollout while validating system performance under operational conditions. Establish data pipelines for continuous environmental data integration and automated literature review updates, ensuring the system remains current with evolving research and changing environmental conditions.

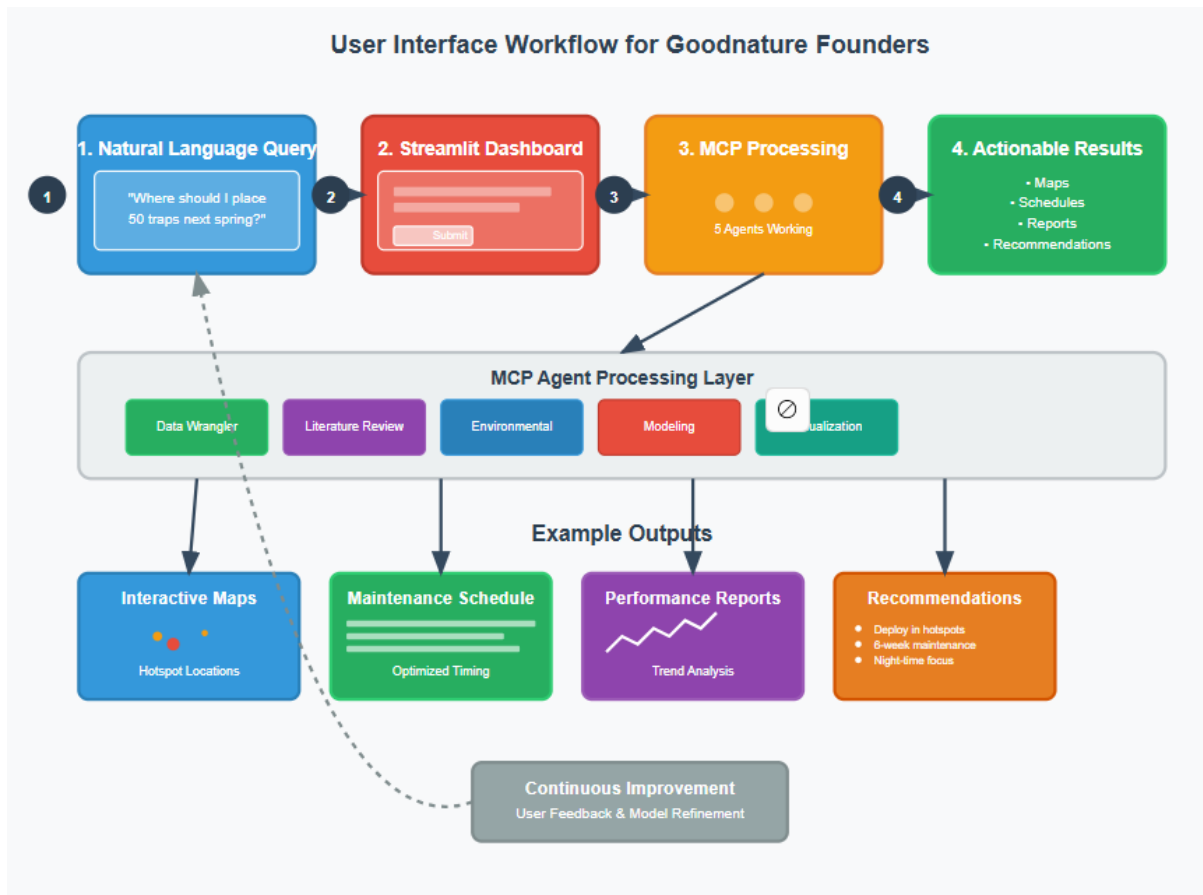


Figure 2: Example user workflow for Goodnature founders. Users submit natural language queries through a lightweight dashboard (e.g., Streamlit). The MCP orchestrator routes requests through agents and returns results as maps, schedules, or reports.

Strategic Resource Allocation

Redirect resources from uniform trap distribution toward concentrated deployment in verified hotspots during optimal environmental conditions (10-20°C temperature range, 60-80% humidity). This approach could potentially achieve equivalent pest control outcomes with 40% fewer devices, freeing resources for expansion into new areas or enhanced monitoring of critical conservation zones.

Continuous Improvement Framework

Establish feedback loops between operational outcomes and model refinement, using new trap performance data to continuously improve predictive accuracy. Implement A/B testing protocols for trap placement and maintenance strategies, allowing empirical validation of model recommendations. Schedule quarterly literature review updates to incorporate new research findings into operational guidance.

Future Development Path

The MCP architecture provides a foundation for expanding analytical capabilities.

Priority enhancements include integration with additional environmental data sources (satellite imagery, ecological surveys), development of natural language query interfaces for non-technical staff, and expansion of predictive capabilities to include pest population dynamics and ecosystem impact assessment. Further development in areas such as a Biodiversity Impact Agent and/or Community Engagement Agent might be a way forward as the architecture grows.

This investigation demonstrates that effective analytics for conservation technology requires integration of multiple data sources, domain expertise, and adaptive systems architecture. The MCP framework provides Goodnature with both immediate operational improvements and a platform for continuous innovation in evidence-based conservation. The key insight is that successful analytics is not about achieving perfect predictions, but about building systems that improve decision-making under uncertainty while remaining adaptable to changing conditions and new information.

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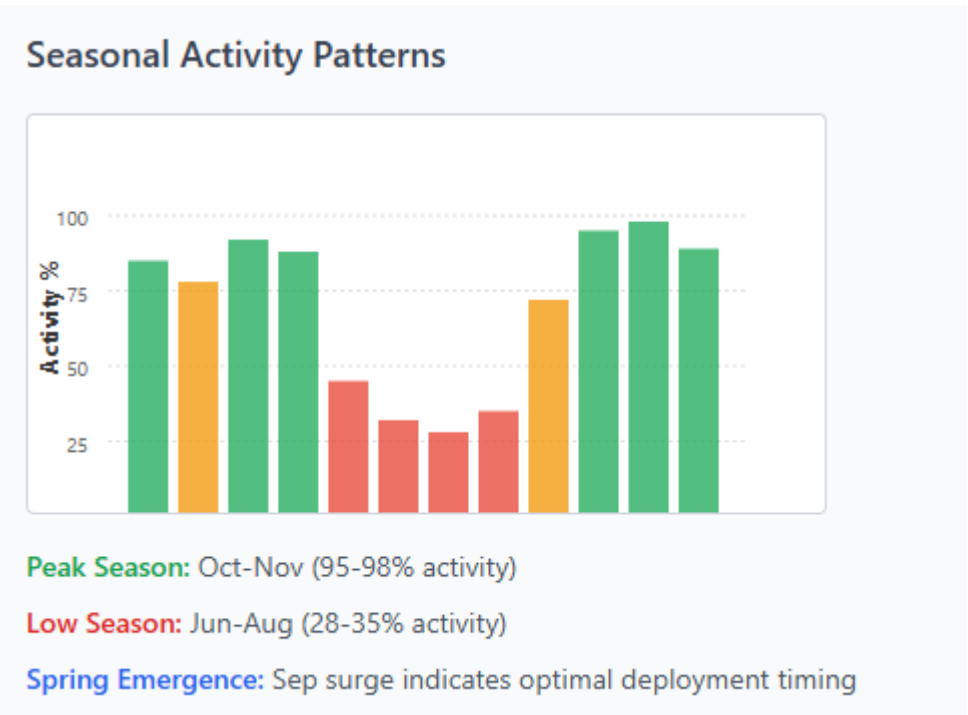
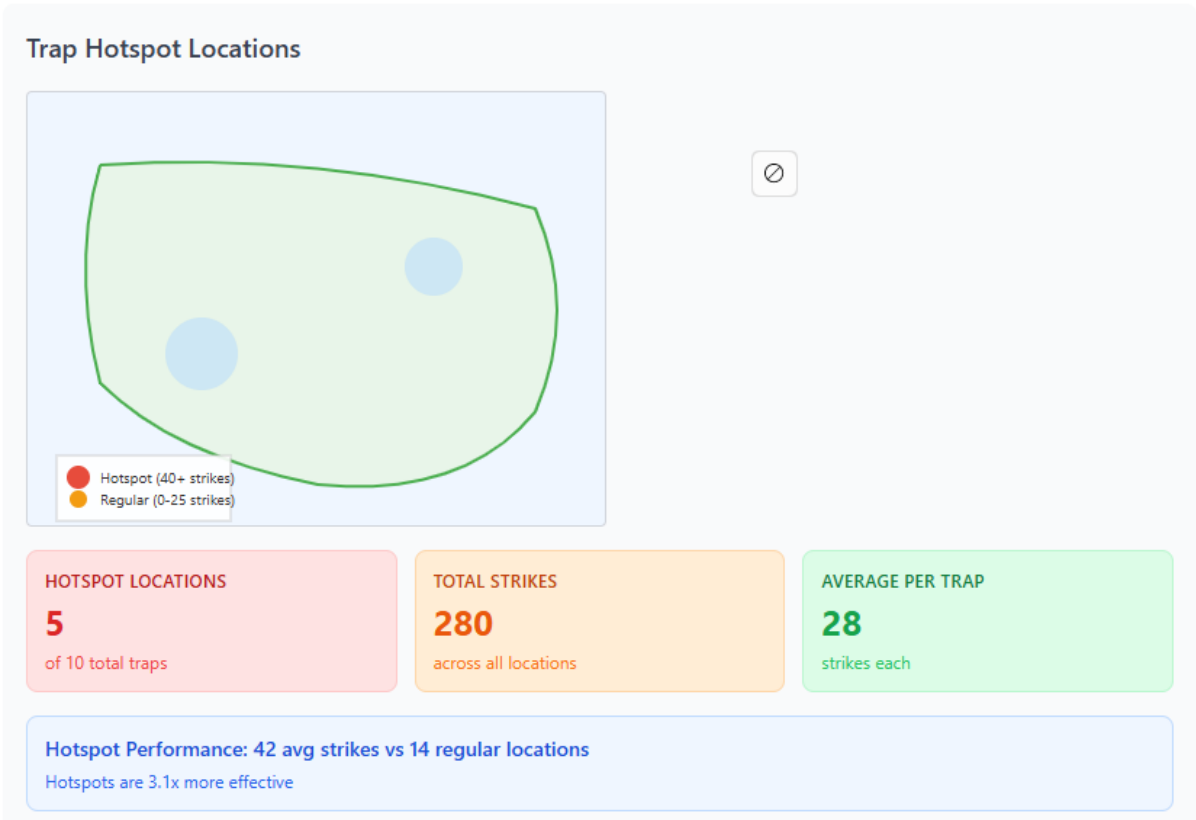
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Appendix A: Prototype Interactive Dashboard

The following excerpt demonstrates a working React-based dashboard developed as a proof-of-concept interface for Goodnature’s MCP system. It illustrates how non-technical users could view hotspots, seasonal activity, and receive automated recommendations in real-time.

Goodnature Trap Performance Dashboard



MCP System Recommendations

Immediate Actions:

- Deploy 15 additional traps in Rimutaka Range hotspot (51 strikes)
- Schedule maintenance for Kapiti Coast (6 weeks overdue)
- Prioritize Zealandia area during Oct-Nov peak season
- Focus 70% of monitoring resources on 5 hotspot locations

Strategic Planning:

- Reduce winter monitoring frequency by 40% (Jun-Aug)
- Implement weather-based deployment timing
- Establish performance-based maintenance schedules
- Target 40% resource savings through hotspot concentration

Projected Impact: Concentrating resources on 5 hotspots could achieve equivalent pest control with 50% fewer devices, freeing resources for 3 new strategic locations.